

The formula with irrational roots

WORKSHEET



Interactive version

Name _____

Class _____

Date _____

REMEMBER

Square roots and the formula

Level 1 • threshold

1.Which is $\sqrt{48}$ in its simplest form?

☐ A $16\sqrt{3}$

☐ B $2\sqrt{12}$

☐ C $6\sqrt{8}$

☐ D $4\sqrt{3}$

Level 1 • threshold

2.Solve $2x^2 - 5x + 2 = 0$ with the formula.

Answer: _____

Level 1 • threshold

3.Solve $(x + 1)^2 = 6$. Write the roots exactly: type $\sqrt{}$ as sqrt and $\pm$ as +-, or give the two roots separated by a semicolon.

Answer: _____

WORKED EXAMPLES

Worked examples

YOUR TURN

Level 2 • satisfactory

Solve $x^2 + 2x - 7 = 0$. Give the roots exactly, in their simplest form.

Answer: _____

Example 2 • When $a \neq 1$: $2x^2 - 6x + 3 = 0$

COEFFICIENTS

$$a = 2, b = -6, c = 3.$$

DISCRIMINANT

$$D = (-6)^2 - 4 \cdot 2 \cdot 3 = 36 - 24 = 12, \text{ and } \sqrt{12} = 2\sqrt{3}.$$

FORMULA

$$x_{1,2} = \frac{6 \pm 2\sqrt{3}}{2 \cdot 2} = \frac{6 \pm 2\sqrt{3}}{4}.$$

SIMPLIFY • YOUR TURN

ROUND, IF ASKED • YOUR TURN

YOUR TURN

Level 2 • satisfactory

Solve $3x^2 - 4x - 2 = 0$. Give the roots exactly, in their simplest form.

Answer:

CHECK

Check yourself

Level 1 • threshold

1.

Simplify $\frac{6 \pm \sqrt{12}}{2}$.

(A) $3 \pm \sqrt{12}$

(B) $3 \pm \sqrt{3}$

(C) $3 \pm \sqrt{6}$

(D) $3 \pm 2\sqrt{3}$

Level 2 • satisfactory

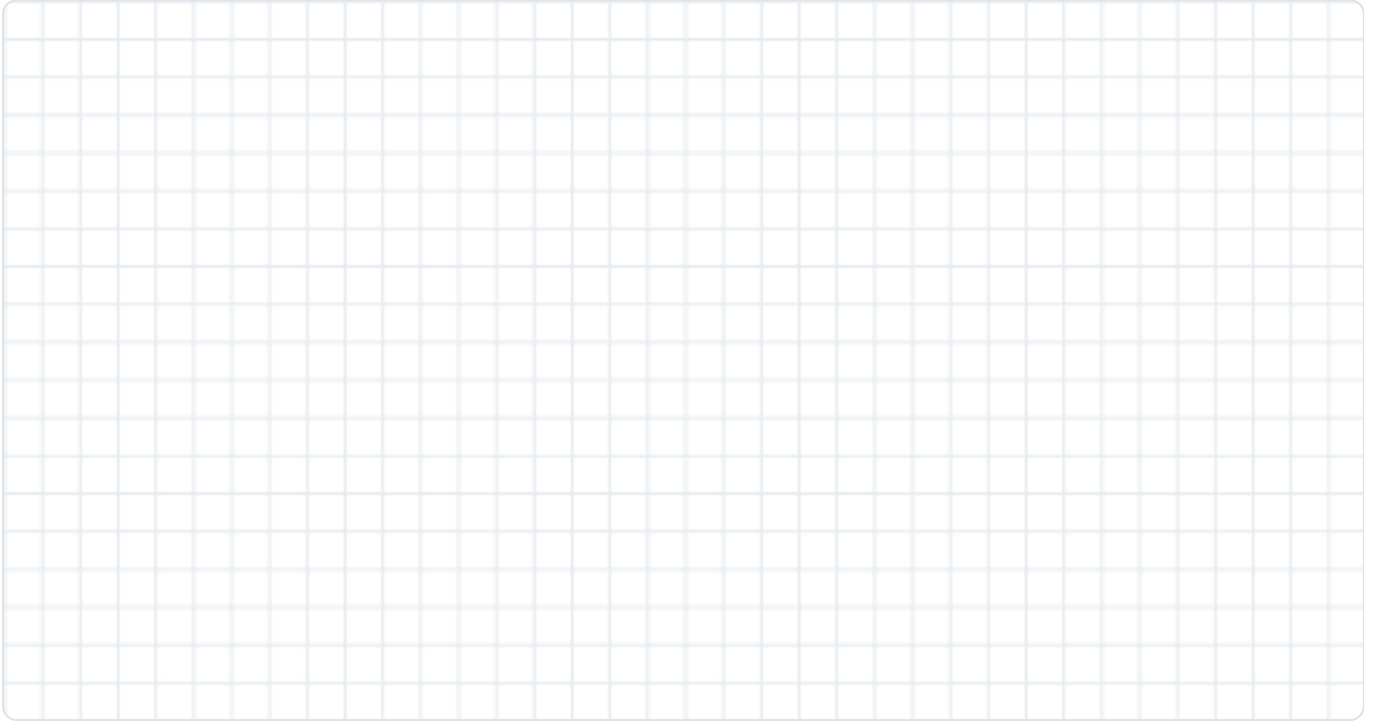
2.

Solve $x^2 - 2x - 2 = 0$. Give the roots exactly.

Answer:

3.

Solve $4x^2 - 4x - 1 = 0$. Show your working, and write the roots exactly, in their simplest form.

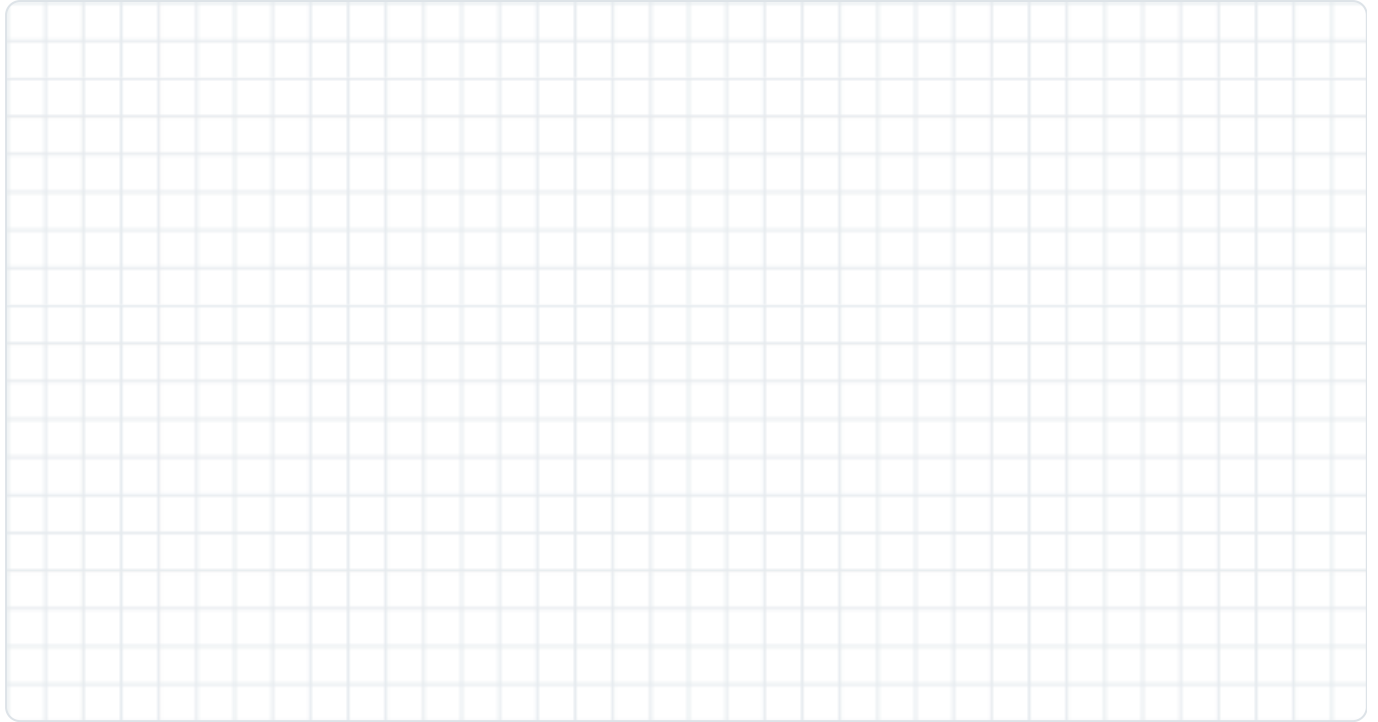
**4.**

Solve $x^2 + 8x + 10 = 0$. Give the roots exactly.

Answer:

5.

Above, you found that $x = 1 + \sqrt{3}$ is a root of $x^2 - 2x - 2 = 0$. Show it by substituting: work out the left side exactly, without a calculator.



EXAM TASK

Exact roots without a calculator

Level 3 · basic

PUPP style · 3 points

The equation $x^2 - 10x + 18 = 0$ is given.

- Show that its discriminant is 28.
- Show that its roots are $5 - \sqrt{7}$ and $5 + \sqrt{7}$.
- Between which two consecutive whole numbers does each root lie? Explain without a calculator.

